

MRT4Depth: Metamorphic Robustness Testing for Ground-Truth-Free Evaluation of Monocular Depth Estimation Models

Patience Chew Yee Cheah , Boon Giin Lee , Senior Member, IEEE, Dave Towey , Senior Member, IEEE, David Chieng , Senior Member, IEEE, Tsong Yueh Chen , Life Fellow, IEEE, and Zhi Quan Zhou , Member, IEEE

Abstract—Monocular depth estimation (MDE) must maintain stable predictions under minor input perturbations to ensure robustness and reliability, particularly in safety-critical applications. Existing robustness-evaluation methods rely on comparisons with ground-truth depth data, which are costly to obtain and often unavailable. This limitation constitutes an instance of the oracle problem, where expected outputs for test inputs cannot be determined. To address this challenge, this study introduces an MRT-based framework for evaluating the robustness of MDE models without requiring ground-truth depth annotations (MRT4Depth). Prediction consistency is measured using both pixelwise consistency and pairwise ordinal depth consistency, the latter being specifically designed to capture the relative ordering property inherent in depth estimation. The proposed framework is applied to evaluate four state-of-the-art MDE models, including UniDepthV2, Depth Pro, ZoeDepth, and Depth Anything 3 on 4 benchmark datasets covering indoor and outdoor scenes, using 18 common image corruptions and four Euclidean transformations. The results indicate that the evaluated models exhibit substantially different relative robustness characteristics across perturbation types. The findings further reveal that high prediction accuracy and stable accuracy under perturbations do not necessarily correspond to high prediction consistency. In general, MRT4Depth enables ground-truth-free robustness evaluation and reveals aspects of prediction consistency that accuracy-based robustness metrics do not capture.

Index Terms—Evaluation method, metamorphic robustness testing (MRT), metamorphic testing (MT), monocular depth estimation (MDE), robustness, sensitivity.

I. INTRODUCTION

MONOCULAR depth estimation (MDE) is a computer vision task that infers scene depth from a single

image. Advances in generalization performance of deep-learning-based MDE models in diverse environments have enabled their widespread adoption in applications, such as autonomous driving [1], robotics [2], drones [3], and endoscopic surgery [4]. In real-world deployment, these models encounter natural input perturbations arising from the image formation process, including environmental conditions, sensor noise, and rendering variations [5]. Strong performance in clean and diverse datasets does not guarantee reliable behavior under such perturbations [6]. Ensuring reliable operation therefore requires MDE models to satisfy *robustness*, a fundamental requirement of trustworthy artificial intelligence systems [7].

Robustness receives varied interpretations in MDE studies, including accuracy in adverse weather and lighting conditions [8], resilience to image corruptions [9], [10], and resistance to adversarial attacks [11]. In the broader machine learning literature, robustness is formally defined as the ability of a *robustness target* to remain stable under interventions on a *modifier*, within a specified *modifier domain* and *tolerance* [12]. By defining the robustness target as the predictions of an MDE model and the modifier as naturally occurring nonadversarial input perturbations, robustness is characterized as prediction consistency under such perturbations. Evaluating whether an MDE model satisfies this robustness requirement requires an appropriate assessment method.

Existing MDE robustness-evaluation benchmarks [6], [9] assess robustness by comparing predicted depths with ground-truth depth data. However, acquiring accurate ground-truth depth maps remains costly and is often impractical [13]. This limitation corresponds to the oracle problem in software testing [14], in which the expected outputs for the test inputs are not available. As a result, current robustness-evaluation approaches remain largely confined to labeled datasets, limiting their applicability to unlabeled or domain-specific data. Furthermore, most existing evaluations emphasize accuracy stability under input perturbations and overlook prediction consistency [15]. Even when accuracy remains stable, inconsistent depth predictions can undermine reliability, particularly in safety-critical applications, such as autonomous driving [16]. This concern is especially significant when the same static scene produces different depth predictions exclusively due to image-level perturbations, as occurs in

Received 28 January 2026; revised 15 May 2026; accepted 7 July 2026. Date of publication 13 July 2026; date of current version 24 July 2026. Associate Editor: I. Ahmed. (Corresponding author: Boon Giin Lee.)

Patience Chew Yee Cheah, Boon Giin Lee, Dave Towey, and Zhi Quan Zhou are with the School of Computer Science, University of Nottingham Ningbo China, Ningbo 315100, China (e-mail: patiencechewyee.cheah@nottingham.edu.cn; boon-giin.lee@nottingham.edu.cn; dave.towey@nottingham.edu.cn; George.Zhou@nottingham.edu.cn).

David Chieng is with the Department of Electrical and Electronic Engineering, University of Nottingham Ningbo China, Ningbo 315100, China (e-mail: david.chieng@nottingham.edu.cn).

Tsong Yueh Chen is with the Department of Computing Technologies, Swinburne University of Technology, Hawthorn, VIC 3122, Australia (e-mail: tychen@swin.edu.au).

Digital Object Identifier 10.1109/TR.2026.3711986

systems operating at high frames per second where geometric changes between consecutive frames are minimal. These limitations highlight the need for a robustness-evaluation method that assesses prediction consistency while addressing the oracle problem.

Metamorphic testing (MT) provides a software testing paradigm designed to address the oracle problem [17]. MT relies on metamorphic relations (MR), which define the expected relationships among multiple inputs and their corresponding outputs [18]. Violations of these relations indicate incorrect system behavior. Metamorphic robustness testing (MRT) [19] extends this paradigm by explicitly focusing on robustness properties rather than functional correctness. MT and MRT have been applied to various computer-vision tasks, including object detection [20] and image segmentation [21], to identify faulty behaviors and assess robustness. However, the application of MRT to MDE remains limited. Unlike other vision tasks, MDE produces dense, continuous-valued depth maps [22], [23], which require a different and specialized set of MRs adapted to depth prediction characteristics.

This work introduces an MRT-based framework for evaluating the robustness of MDE models without requiring ground-truth depth annotations (MRT4Depth), thus addressing the oracle problem in MDE robustness evaluation. The main contributions of this work are summarized as follows.

- 1) Propose MRT4Depth as a ground-truth-free robustness-evaluation framework for MDE, an area where MRT remains underexplored. The framework supports robustness assessment on unlabeled datasets representative of target deployment domains.
- 2) Define 22 MRs based on prediction consistency under 18 image corruptions and four Euclidean transformations, supported by mathematical justifications. The prediction consistency is measured using the mean absolute difference and pairwise ordinal consistency, where the latter captures the relative ordering property of depth estimation.
- 3) Present a comparative robustness study of four state-of-the-art MDE models under the proposed 22 MRs. The results show that prediction accuracy and accuracy stability do not necessarily imply prediction consistency and that relative robustness differs substantially among models and perturbation types.

The rest of this article is organized as follows. Section II reviews related work on MDE, robustness evaluation, and MRT. Section III presents the MRT4Depth method, including the defined MRs, and describes the datasets and MDE models used in this study. Section IV reports and discusses the experimental results and outlines the limitations of the study. Finally, Section V concludes this article.

II. RELATED WORK

This section reviews existing studies related to MDE, robustness evaluation of MDE models, and MRT. The discussion highlights the gap between strong generalization performance and robustness guarantees in existing MDE models, highlights

limitations of current robustness-evaluation practices, and motivates the adoption of MRT to address the oracle problem in MDE robustness evaluation.

A. MDE Models

MDE aims to predict a dense depth map per-pixel from a single RGB image, where each depth value represents the distance along the optical axis of the camera to the corresponding point in the scene. These depth values are continuous and typically expressed in physical units, such as meters. Eigen et al. [22] formulated MDE as a pixelwise regression problem and introduced an early deep-learning-based approach, establishing a foundation for subsequent research.

Recent state-of-the-art MDE models, including ZoeDepth [24], Depth Pro [25], UniDepthV2 [26], and Depth Anything 3 [27], have achieved high accuracy and strong generalization performance, even in scenes not observed during training. These models adopt transformer-based architectures and rely on supervised learning with large-scale, multidataset training. Xu et al. [28] attributed this generalization capability to the scaling of model sizes to billions of parameters and the use of increasingly diverse and extensive training datasets.

Despite these advances, strong generalization on clean and diverse datasets does not imply robustness under input perturbations. Xian et al. [6] showed that MDE models with high generalization performance may still exhibit a substantial vulnerability to image corruptions. Although robustness constitutes a critical requirement for reliable real-world deployment, relatively few studies have systematically explored the robustness of deep-learning-based MDE models.

B. Robustness Evaluation of MDE Models

Several studies have introduced robustness-evaluation benchmarks for MDE models under nonadversarial perturbations. Inspired by ImageNet-C [29], which evaluates the robustness in image classification, Xian et al. [6] applied 15 standard image corruptions to six widely used MDE datasets, including IBims-1, NYUv2, and KITTI. Similarly, Kong et al. [9] extended this framework by introducing three additional corruptions and categorizing 18 perturbations into weather and lighting conditions, sensor failures and motion, and data processing problems. In the medical domain, Sharifan et al. [10] proposed the RoDEM benchmark, which focuses on surgical imagery and task-specific perturbations. Beyond 2-D image corruptions, Nugent et al. [15] introduced the procedural depth evaluation (PDE) benchmark, which simulates 3-D scene-level perturbations using synthetic scene generation.

These robustness-evaluation benchmarks rely primarily on comparisons between predicted depth maps and ground-truth depth data and focus on accuracy stability under perturbations. Commonly used metrics include absolute relative error [6], [15] and unified accuracy measures that combine error and threshold-based criteria [9], [10]. However, acquiring accurate pixelwise ground-truth depth maps remains time-consuming and often requires expensive and specialized hardware [13]. Moreover, some datasets provide ground-truth depth maps with lower

spatial resolution than their corresponding RGB images [30], further limiting the fidelity of the evaluation. These challenges give rise to the oracle problem, where expected outputs for test inputs remain unavailable.

In addition, existing robustness evaluations largely emphasize accuracy stability, typically measured as changes in prediction error between original and perturbed inputs. Such metrics overlook prediction consistency, which quantifies the difference between a model's outputs on clean and perturbed inputs. Prediction consistency represents a stricter robustness criterion than accuracy stability, as perfectly consistent predictions imply stable accuracy, while stable accuracy does not guarantee consistent predictions [15]. In general, most existing MDE robustness-evaluation methods depend on ground-truth depth data and provide limited information on prediction consistency.

C. Robustness Evaluation of MT

The authors in [31] and [32] invented MT that alleviates the oracle problem by validating system behavior through MRs rather than explicit expected outputs, as also described by Segura et al. [17]. MRs define expected relationships among multiple inputs and their corresponding outputs, and violations of these relations indicate faulty behavior. Zhou et al. [19] extended this concept to MRT, which explicitly targets robustness properties instead of functional correctness. Li et al. [33] emphasized the role of MRs in guiding both the generation of the test input and the verification of the output.

Motivated by the oracle problem, several studies have applied MT and MRT to computer-vision systems. Existing work evaluated object detection [20], [34], image segmentation [21], pose estimation [35], and autonomous driving systems [36], [37] by assessing output consistency under image transformations, such as 2-D image-plane transformations, weather effects, and lighting variations. A common assumption in these studies is that robust models should maintain consistent outputs under input perturbations. In addition, Wang and Su [34] and Tian et al. [36] emphasized that effective MRs should reflect realistic deployment conditions.

Despite these advances, applications of MRT to MDE remain limited. MDE models produce dense depth maps with continuous-valued outputs at every pixel, which differ fundamentally from the discrete or sparse outputs of many other vision tasks. This characteristic necessitates a specialized set of MRs adapted to depth prediction. To address the oracle problem in MDE robustness evaluation, this study adopts MRT and designs a set of MRs based on nonadversarial perturbations to assess prediction consistency without relying on ground-truth depth data.

III. METHODS

Fig. 1 illustrates the overview of the proposed MRT4Depth framework. Here, let M denote the MDE model under test. The *robustness target*, defined as the depth map produced by the model M , is used in an evaluation that focuses on the consistency of the predicted depth values rather than on absolute accuracy or performance stability. For each test case, a follow-up image

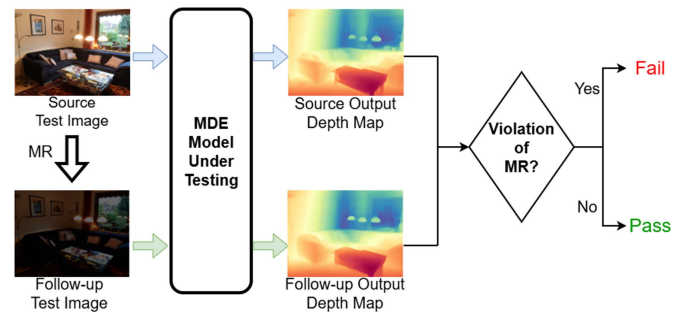


Fig. 1. Overview of the proposed MRT4Depth framework.

I' is generated from a source image I according to an MR. An MR functions as a *modifier* by perturbing the input within a specific *domain* or parameter range. The framework compares the two depth maps produced by M from I and I' . A test fails when the outputs violate the MR beyond the *tolerance* allowed, indicating an insufficient *robustness target* with respect to the *modifier*. MRT4Depth operates without requiring ground-truth depth data.

In MRT4Depth, MRs serve as *modifiers* that perturb a source image I to generate a follow-up image I' , while also allowing consistency checks between the corresponding depth outputs $M(I)$ and $M(I')$. Although real-world perturbations are diverse, the proposed framework focuses on common image corruptions and Euclidean transformations that preserve the underlying scene geometry and do not represent physical changes in camera viewpoint. Under these conditions, depth predictions are expected to remain consistent. Thus, MRs are formulated on the basis of these two categories of transformations and the resulting depth consistency. When these conditions do not hold, depth consistency can no longer be assumed, and alternative MRs become necessary. The rationale for the expected depth consistency under image corruptions and Euclidean transformations is presented below.

A. MRs for Image Corruptions

This study utilized the image corruptions introduced in RoboDepth [9], which includes 18 common image corruptions that simulate natural variations in image appearance while preserving the underlying scene geometry and camera viewpoint. The 18 corruption-related MRs follow the RoboDepth taxonomy as follows.

- 1) *Lighting and weather corruptions: Brightness, dark, fog, frost, snow, and contrast.* These perturbations alter image intensity and color without affecting scene geometry, hence the predicted depths should remain consistent. Although *frost* and *snow* introduce occluding artifacts, robust models are expected to rely on global scene structure and preserve prediction consistency. Therefore, these cases assess robustness rather than correctness.
- 2) *Camera malfunction and movement corruptions: Defocus blur, glass blur, motion blur, zoom blur, elastic transformation, and color quantization.* These perturbations

distort edges and textures that should not affect the predicted depths. Although elastic transformation locally deforms pixel coordinates, robust models are expected to infer scene geometry from global structure and maintain prediction consistency.

- 3) *Data collection and transmission corruptions: Gaussian noise, impulse noise, shot noise, ISO noise, pixelation, and JPEG compression.* These corruptions introduce noise and compression artifacts without altering scene predicted depths.

Here, let I denote an image of height H and width W , and let D denote its corresponding true depth map with the same spatial dimensions. Let the image domain Ω represent the set of all the pixel coordinates in both I and D

$$\Omega = \{(u, v) \in \mathbb{Z}^2 : 1 \leq u \leq H, 1 \leq v \leq W\}. \quad (1)$$

Let Π denote the projection function that maps a real-world 3-D scene point to an image coordinate. Let $\mathbf{P}$ denote the set of 3-D points in the scene such that, for every pixel $(u, v) \in \Omega$, there exists a corresponding 3-D point $p^{uv} \in \mathbf{P}$ satisfying

$$\mathbf{P} = \{p^{uv} \in \mathbb{R}^3 : \forall (u, v) \in \Omega, \Pi(p^{uv}) = (u, v)\} \quad (2)$$

where $p^{uv} = (x^{uv}, y^{uv}, z^{uv})$ is expressed in the camera coordinate system, and z^{uv} denotes the depth along the camera's Z -axis. The true depth map D is therefore defined as

$$\forall (u, v) \in \Omega, \quad D_{uv} = z^{uv}. \quad (3)$$

Let t_C denote a corruption transformation that affects only the values of the pixel values of the image. Let $I' = t_C(I)$ denote the corrupted image. Transformation t_C is assumed to satisfy the following properties.

- 1) The set of 3-D points $\mathbf{P}$ remains unchanged.
- 2) The projection function Π remains unchanged.
- 3) The correspondence between pixels and 3-D points remains unchanged.
- 4) The pixel coordinate mapping remains identical

$$\forall (u, v) \in \Omega, \quad (u', v') = (u, v) \quad (4)$$

where $(u', v') \in \Omega'$ denotes the pixel coordinates in I' .

Under these assumptions, each pixel (u', v') corresponds to the same 3-D point p^{uv} as in the original image. Thus, the true depth map D' of I' remains identical to D

$$\forall (u, v) \in \Omega, \quad D'_{uv} = z^{uv} = D_{uv} \Rightarrow D' = D. \quad (5)$$

Consequently, a robust model M for $t_C(I)$ should satisfy the approximation within a tolerance τ , defined as

$$I' = t_C(I) \Rightarrow M(I) \approx M(I') \quad (6)$$

where $M(I)$ and $M(I')$ should differ by less than τ . Existing robustness evaluation of MDE measures the accuracy under image corruptions by comparing predictions with ground-truth depth maps, implicitly assuming that $M(I) = D$ and $M(I') = D$. Taking into account scenarios where ground-truth depths are not available, MRT4Depth assumes $M(I) \approx M(I')$. Lastly, this study adopts parameter configurations with five severity levels, consistent with the benchmarks in [6] and [9] for the 18 corruption-related MRs.

B. MRs for Euclidean Transformations

This study considers basic 2-D image-plane Euclidean transformations, including reflection (flip), rotation, and translation. These transformations modify the pixel coordinates while preserving the underlying scene geometry and do not represent physical changes in the camera viewpoint. Composite transformations, such as glide reflection, are excluded because they involve numerous possible combinations. Vertical flipping is also omitted because it reverses the natural top-to-bottom structure of a scene. Existing research has shown that the vertical position of an object in an image provides an important depth cue for MDE [38]. Consequently, vertical flip does not provide a meaningful robustness test for MDE models. The four MR based on Euclidean transformations are defined as follows.

- 1) *Horizontal Flip:* A horizontal flip of the input image should produce a correspondingly flipped depth map.
- 2) *Rotation:* Rotating the input image should produce a correspondingly rotated depth map.
- 3) *Horizontal Translation:* Horizontal shifts of the input image should produce equivalently shifted depth maps.
- 4) *Vertical Translation:* Vertical shifts of the input image should produce equivalently shifted depth maps.

Here, let t_E denote a Euclidean transformation that alters the spatial arrangement of image pixels. For $I' = t_E(I)$, not all pixels in the transformed image Ω' have valid correspondences in Ω . The set of pixels with valid correspondences is defined as

$$\Omega^{\text{valid}} = \{(u, v) \in \Omega : t_E(u, v) \in \Omega'\}. \quad (7)$$

t_E is assumed to satisfy the following properties.

- 1) The set of 3-D points $\mathbf{P}$ remains unchanged.
- 2) The transformation is rigid and preserves Euclidean distances.
- 3) The new projection satisfies $\Pi' = t_E \circ \Pi$.
- 4) Pixel-to-point correspondences remain unchanged.
- 5) Pixel coordinates between I and I' satisfy

$$\forall (u, v) \in \Omega^{\text{valid}}, \quad (u', v') = t_E(u, v). \quad (8)$$

Under these assumptions, each pixel (u', v') corresponds to the same 3-D point p^{uv} , as $t_E(u, v)$. Hence, the new true depth, D' , is obtained by applying t_E to D , defined as

$$\begin{aligned} \forall (u, v) \in \Omega^{\text{valid}}, \quad D'_{t_E(u, v)} &= z^{uv} = D_{uv} \\ &\Rightarrow D' = t_E(D). \end{aligned} \quad (9)$$

Thus, a robust model M for $t_E(I)$ should satisfy the approximation within a tolerance τ for pixels with valid correspondences in both the source and follow-up images (denoted as Ω^{valid}), defined as

$$I' = t_E(I) \Rightarrow t_E(M(I)) \approx M(I') \text{ on } \Omega^{\text{valid}} \quad (10)$$

where $t_E(M(I))$ and $M(I')$ should differ by less than τ . Only t_E that preserve a sufficient proportion of valid pixels are considered. Specifically, transformations must satisfy

$$\frac{|\Omega^{\text{valid}}|}{|\Omega|} \geq \rho. \quad (11)$$

TABLE I
PARAMETER CONFIGURATIONS FOR EACH t_E (HORIZONTAL FLIP HAS NO PARAMETERS)

Transformation	Parameter	Values
Horizontal Flip	-	-
Rotation	Angle ($^\circ$)	-3, -2, -1, 1, 2, and 3
Horizontal Translation	Shift Pixel (pixel)	-30, -20, -10, 10, 20, and 30
Vertical Translation	Shift Pixel (pixel)	-30, -20, -10, 10, 20, and 30

A low value of ρ would allow Euclidean transformations t_E that produce large invalid regions in I' , potentially causing MR violations due to missing image content rather than a actual lack of robustness. In contrast, an excessively high value of ρ would restrict t_E to very small perturbations, making I' nearly identical to I and limiting the effectiveness of the evaluation. Specifically, on IBims [39] dataset, horizontal translations of 50, 100, and 150 pixels (shifting left or right) yielded valid pixel ratios of 92%, 84%, and 77%, respectively. The percentage of MR violations increased sharply between 50 and 100 pixel (28–32 percentage point for left shift and 15–37 percentage point for right shift), but increased only slightly between 100 and 150 pixels (0–19 percentage point for left shift and 12–25 percentage point for right shift). This indicates that $\rho \approx 0.9$ marks a transition point where violations reflect model behavior rather than missing image content. Therefore, $\rho = 0.9$ is selected to ensure that I' retains at least 90% valid pixels while still allowing perturbations of sufficient magnitude for a meaningful robustness evaluation.

Cropping invalid regions would alter the image dimensions, which can affect camera intrinsic estimation in some MDE models [25], [26]. To isolate robustness under t_E without introducing changes in the camera parameter, the original image size is preserved.

To maintain consistency with existing benchmarks that use five severity levels for image corruptions, the same number of parameter settings was adopted for the Euclidean-transformation MRs. Both positive and negative values were included to avoid directional bias. Consequently, six parameter values were selected, consisting of three negative and three positive values. These settings were chosen to satisfy the valid-pixel-ratio defined in (11). Table I lists the parameter configurations for the four Euclidean-transformation MRs. Each transformation includes six parameter values, except horizontal flip, which has no parameters.

Overall, a total of 22 MRs are defined as *modifiers*. A robust MDE model is expected to maintain stable predictions under all perturbations. Although these perturbations alter image characteristics, such as color, texture, and orientation, a robust model should rely on geometrically meaningful features that are invariant to appearance changes [40].

C. Evaluation Metrics and Tolerance

As a continuous per-pixel regression task, MDE naturally produces output variations under input perturbations. Therefore, all MRs enforce approximate equality rather than exact equality, as defined in (6) and (10). The approximation is quantified using

TABLE II
DATASETS USED FOR MDE MODEL EVALUATION

Dataset	#Img	Dimensions	Scene	Intensity	Red	Green	Blue	Noise (%)
IBims-1	100	640×480	In	108±24	115±24	107±25	99±29	0.20±0.08
NYUv2	654	640×480	In	108±28	121±30	103±28	98±31	0.14±0.02
Make3D	133	1704×2272	Out	125±23	123±25	127±22	118±25	0.27±0.11
KITTI	652	1224×370,	Out	94±9	92±10	95±9	92±12	1.91±0.55
		1226×370,						
		1238×374,						
		1241×376,						
		1242×375						

Make3D originally contained 134 images; one corrupted image was removed, leaving 133 images. Scene type: Indoor (In), Outdoor (Out).

the mean absolute difference, Diff, as defined below

$$\text{Diff} = \frac{1}{|\Omega^{\text{valid}}|} \sum_{(u,v) \in \Omega^{\text{valid}}} |t(M(I))_{uv} - M(I')_{uv}| \quad (12)$$

where t denotes the identity function for MRs based on corruption ($t = t_C$) and MRs based on Euclidean-transformation $t = t_E$. An MR violation occurs when $\text{Diff} > \tau$.

Tolerance τ reflects application-specific strictness. Indoor scenes typically exhibit a maximum depth of 10 m (NYUv2 [41]), while outdoor scenes reach approximately 90 m (KITTI [42]). Because depth prediction differences generally scale with scene depth, a fixed τ is not suitable for datasets with different depth ranges. A larger τ allows for greater prediction differences and typically reduces the MR-violation percentage, while a smaller τ imposes a stricter criterion and typically increases it. Section IV-B shows that the robustness rankings of the model remain stable over a range of τ values. Since no specific application is considered, a mid-range tolerance is adopted, that is, 5% of the maximum depth for each type of scene, producing $\tau = 0.5$ for indoor datasets and $\tau = 4.5$ for outdoor datasets.

Although depth maps contain metric values, they also encode ordinal relationships among pixels, where larger values indicate greater distance, and smaller values indicate nearer depth. This property is intrinsic to depth estimation and can therefore be used for evaluation. Since depth maps are expected to remain unchanged under the 22 perturbations, the ordinal relationships between pixel pairs should also be preserved. Consequently, in addition to the consistency measured in pixels by Diff, the ordinal pairwise consistency is also considered. Let i and j denote pixel indices in a depth map of length $N = |\Omega^{\text{valid}}|$. The set of unordered pixel pairs, denoted by Γ , is defined as

$$\Gamma = \{(i, j) | 1 \leq i < j \leq N\}. \quad (13)$$

The pairwise ordinal consistency rate, denoted by Ord, is defined as the proportion of concordant pixel pairs whose relative ordering is preserved

$$\begin{aligned} \text{Ord} &= \frac{1}{|\Gamma|} \sum_{(i,j) \in \Gamma} \mathbb{I}(\text{sign}(t(M(I))_i - t(M(I))_j) \\ &= \text{sign}(M(I')_i - M(I')_j)) \end{aligned} \quad (14)$$

where $\mathbb{I}(\cdot)$ denotes the indicator function, which returns 1 if its argument is true and 0 otherwise. The function $\text{sign}(x)$ returns 1 if $x > 0$, -1 if $x < 0$, and 0 if $x = 0$.

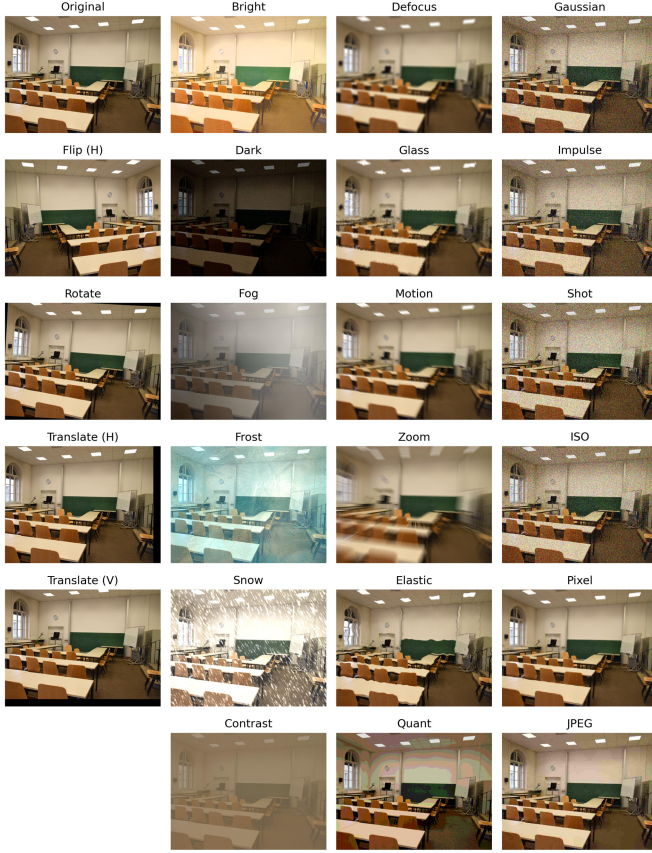


Fig. 2. Illustration of an original IBims-1 image and the corresponding 22 follow-up images generated using each MR. The perturbation severity corresponds to level 3 for corruption-based MRs, -3° for rotation, and -30 pixels for translations.

However, direct computation of Ord requires evaluation of all $\frac{N(N-1)}{2}$ pairs of pixels and is therefore impractical for large images and large-scale experiments. To avoid explicit pairwise enumeration, Ord is approximated using Kendall's Tau K_τ , which is defined as

$$K_\tau = \frac{n_C - n_D}{n_C + n_D} \quad (15)$$

where n_C and n_D are the numbers of concordant and discordant pairs, respectively. Since

$$\text{Ord} = \frac{n_C}{n_C + n_D}. \quad (16)$$

It follows directly that:

$$\text{Ord} = \frac{K_\tau + 1}{2}. \quad (17)$$

Thus, Ord is computed using the standard implementation of Kendall's Tau for efficient evaluation.

The value of Ord lies in $[0,1]$, where 0 indicates that ordinal relationships are not preserved and 1 indicates complete preservation. An MR violation occurs when $\text{Ord} < \theta$. Unlike τ , a fixed threshold θ can be applied to all datasets because Ord is normalized. Section IV-B shows that the robustness rankings

TABLE III
PERCENTAGE (%) OF VALID PIXELS IN FOLLOW-UP IMAGES FOR EACH DATASET UNDER EUCLIDEAN TRANSFORMATIONS: *ROTATION*, *HORIZONTAL TRANSLATION*, AND *VERTICAL TRANSLATION*

Transformation	Param.	IBims	NYUv2	Make3D	KITTI
Rotation	-3°	97.6	97.6	97.5	95.6
	-2°	98.4	98.4	98.3	97.1
	-1°	99.3	99.3	99.2	98.6
	1°	99.3	99.3	99.2	98.6
	2°	98.4	98.4	98.3	97.1
	3°	97.6	97.6	97.5	95.6
Translation (H)	-30	93.8	93.8	98.7	92.0
	-20	95.8	95.8	99.1	94.6
	-10	97.9	97.9	99.6	97.3
	10	97.9	97.9	99.6	97.3
	20	95.8	95.8	99.1	94.6
	30	93.8	93.8	98.7	92.0
Translation (V)	-30	95.3	95.3	98.2	97.6
	-20	96.9	96.9	98.8	98.4
	-10	98.4	98.4	99.4	99.2
	10	98.4	98.4	99.4	99.2
	20	96.9	96.9	98.8	98.4
	30	95.3	95.3	98.2	97.6

Mean percentages are reported for the KITTI Dataset, which consists of images with varying dimensions

of the model remain generally stable over a range of θ values. Therefore, a mid-range value, $\theta = 0.95$, is adopted.

Although fixed tolerances τ and θ are used in this study, these values should be selected according to application requirements. Safety-critical applications, such as autonomous driving and robotic navigation, require smaller τ and larger θ for stricter evaluation, while less critical applications, such as novel view synthesis and augmented reality, may allow larger τ and smaller θ .

D. Datasets

The experimental evaluation used four MDE benchmark datasets: NYUv2 [41], IBims-1 [39], KITTI [42] and Make3D [30]. Table II summarizes the main characteristics of the datasets, including the number of images, image dimensions, scene type, mean intensity, mean RGB channel values, and estimated noise level. Only the standard test splits for each dataset were used, ensuring that no test images appeared in the corresponding training sets. All images were processed in their original dimension without resizing during the generation or evaluation of the follow-up test images. Each source image was subjected to 22 perturbation types with multiple parameter settings, resulting in a total of 167 751 test images. Fig. 2 illustrates representative follow-up images generated from a sample IBims-1 image. Table III showed the proportion of valid pixels in each follow-up image I' for Euclidean transformations involving rotation and translation. All transformed images retained more than 90% valid pixels, satisfying the overlap constraint defined in (11).

E. Monocular Metric Depth Estimation Models

The evaluation considered four state-of-the-art monocular metric depth estimation foundation models published within

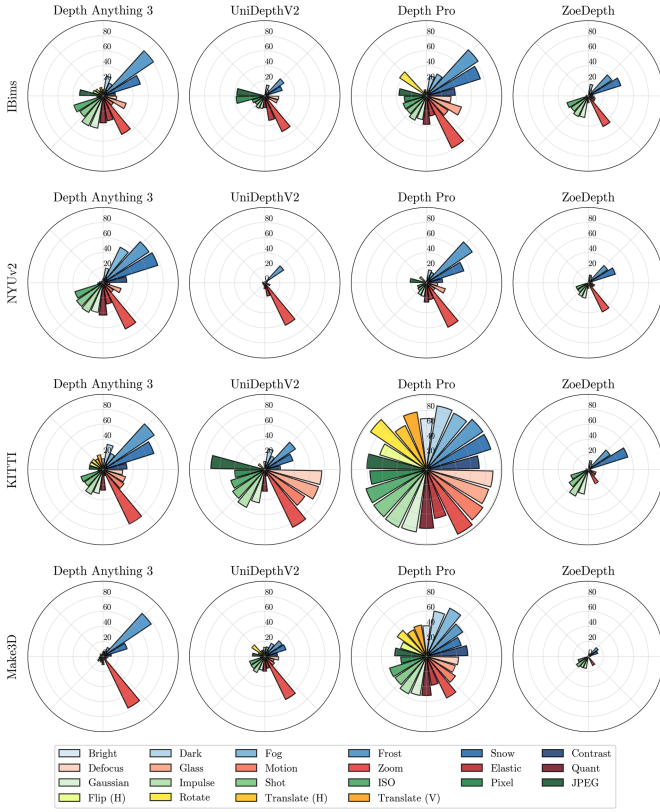


Fig. 3. MR violation (%) measured by Diff under 22 input perturbations for four MDE models on four test datasets.

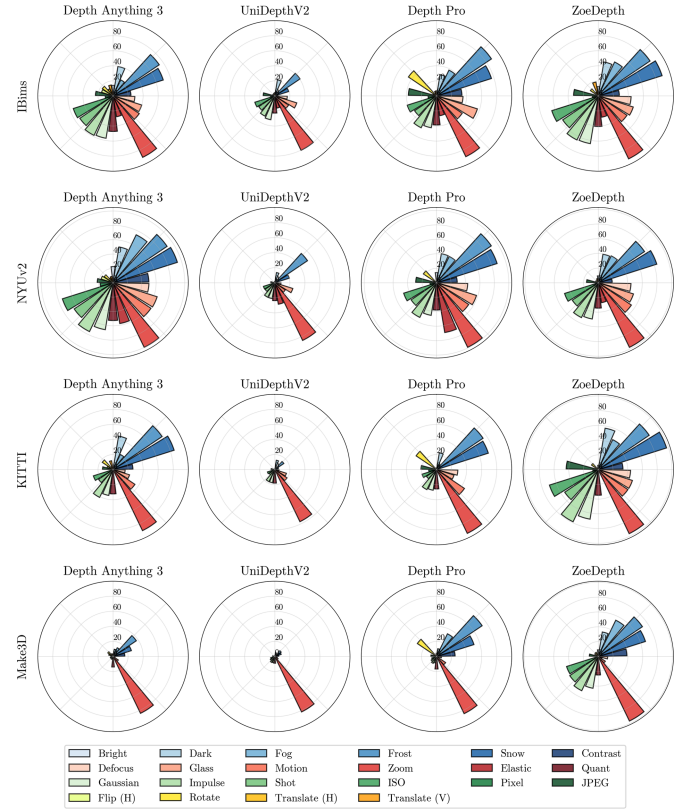


Fig. 4. MR violation (%) measured by Ord under 22 input perturbations for four MDE models on four test datasets.

the last three years: UniDepthV2 (ViT-L model) [26], Depth Pro [25], ZoeDepth (Zoe_NK model) [24], and Depth Anything 3 (DA3NESTED-GIANT-LARGE model) [27]. These models predict absolute metric depth, exhibit strong cross-dataset generalization, and require only a single RGB image as input, without camera-specific parameters.

IV. RESULTS AND DISCUSSION

A. MR-Violation Percentage of MDE Models Under Test

Figs. 3 and 4 show the MR-violation percentages, measured by Diff and Ord, for four MDE models under 22 input perturbations. The results indicated distinct violation patterns by perturbation type and dataset. For example, on IBims, Depth Anything 3 produced low violation rates under *fog* (19% for Diff and 24% for Ord), while the corresponding rates on NYUv2 were much higher (52% for Diff and 74% for Ord), despite the use of the same tolerance τ for both indoor datasets. Similarly, on Make3D, ZoeDepth showed violation rates of approximately 13% (Diff) and 67% (Ord) under *snow*, but the rates increased to approximately 55% (Diff) and 97% (Ord) on KITTI. Since the datasets differ in their characteristics (see Table II), the same model can exhibit different levels of prediction consistency under the same perturbation. For instance, *snow* introduces white pixels that may disrupt local image structure more severely in darker datasets, such as KITTI (mean color intensity of 94), than in brighter datasets, such as Make3D (mean color intensity

of 125). This may partially explain the difference observed on ZoeDepth, although further analysis is needed to determine the exact cause.

Because each perturbation was operated under different units, parameter ranges, and visual effects, direct comparisons across heterogeneous perturbations lack interpretability. For instance, models generally showed higher MR-violation percentages measured by both Diff and Ord under *zoom blur* than under other blur types. This observation does not imply greater robustness to other blur perturbations, as the parameter configuration of *zoom blur* introduces more severe visual distortions, which increases the difficulty of maintaining consistent predictions. Nevertheless, a high MR-violation percentage under a given perturbation indicates insufficient robustness with respect to that perturbation under the specified test configuration and tolerance threshold. If a visually similar perturbation occurs during real-world deployment, the corresponding depth predictions are likely to become unreliable. Such inconsistencies may arise from inherent model limitations or from reliance on visual features that perturbations significantly alter. In either case, high MR-violation percentages indicate that the affected model cannot produce reliable depth maps under those conditions in operational environments.

Although MR-violation percentages do not support direct comparisons across heterogeneous perturbations, they provide a meaningful basis for comparing relative robustness among models. Table IV shows the p -values from two-sided Wilcoxon

TABLE IV
p-VALUES FROM TWO-SIDED WILCOXON SIGNED-RANK TESTS

		ZoeD vs DPro	ZoeD vs Uni2	ZoeD vs DA3	Uni2 vs DPro	Uni2 vs DA3	DPro vs DA3
<i>Diff</i>	IBims	<0.0001	<u>0.0902</u>	<0.0001	<0.0001	<0.0001	0.0004
	NYUv2	<0.0001	<u>0.1423</u>	<0.0001	<0.0001	<0.0001	<0.0001
	KITTI	<0.0001	<0.0001	<0.0001	<0.0001	<u>0.0382</u>	<0.0001
	Make3D	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
<i>Ord</i>	IBims	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<u>0.2894</u>
	NYUv2	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
	KITTI	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.0024
	Make3D	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001

Six pairwise comparisons were conducted between the MR-violation percentages of ZoeDepth (ZoeD), Depth Pro (DPro), UniDepthV2 (Uni2) and Depth Anything 3 (DA3), measured on *Diff* and *Ord* metrics. The results with $p > \alpha = 0.0083$ are underlined, indicating that there were no statistically significant differences between the corresponding pair of models.

signed-rank tests conducted on paired MR-violation percentages. This nonparametric test suits paired numerical data without distributional assumptions. Because six pairwise comparisons occur between four models, the Bonferroni-corrected significance threshold was set to $\alpha = 0.05/6 = 0.0083$. Values below this threshold indicate statistically significant differences that are unlikely to arise from random variation.

Among the evaluated models, Depth Pro produced the highest MR-violation percentages under *Diff* for all perturbations, with statistically significant differences from the other three models ($p < 0.0083$), indicating the lowest pixelwise prediction consistency. One possible explanation is its emphasis on sharp boundary prediction [25], which can amplify pixel-level differences under perturbations and, in turn, increase the MR violations measured by *Diff*. In contrast, ZoeDepth produced the lowest MR-violation percentages under *Diff*, indicating the strongest pixelwise prediction consistency. Its differences from the other models were statistically significant ($p < 0.0001$), except for the comparison with UniDepthV2 on indoor datasets ($p > 0.0083$). Indoor datasets generally have narrower depth ranges, which can limit prediction deviations and reduce observable consistency differences. Outdoor datasets, by contrast, involve much larger depth ranges and may therefore magnify prediction inconsistency. These findings partially agreed with existing work on 3-D scene-level manipulations [15], which also reported lower pixelwise prediction consistency for Depth Pro than for UniDepthV2.

Despite producing the lowest MR-violation percentages under *Diff*, ZoeDepth showed the highest violations rates under *Ord*, with statistically significant differences from the other models ($p < 0.0001$). In contrast, UniDepthV2 produced the lowest MR-violation percentages under *Ord*, indicating the strongest pairwise ordinal consistency, and differed significantly from the other models ($p < 0.0001$).

To further investigate the discrepancy between the MR-violation percentages measured by *Diff* and *Ord*, the depth differences of discordant pairs in both $M(I)$ and $M(I')$ were analyzed. Although Kendall's Tau provides an efficient estimate of *Ord*, it does not identify individual discordant pairs or their associated depth differences. Explicit pairwise enumeration provides such information, but is computationally impractical. Therefore, the analysis was restricted to the smallest dataset,

IBims, and to four representative perturbations, one from each category: *bright*, *elastic*, *pixelate*, and *vertical translation*. Moderate severity was used, i.e., level 3 for corruptions and ten pixels for translation.

In these settings, ZoeDepth showed the highest proportion of pixel pairs in $M(I)$ with depth differences below 0.5 m, exceeding the other models by more than 1.36 percentage points. For computational efficiency, 20% of valid pixels were randomly sampled to approximate the depth-difference distribution of discordant pairs. For all models, more than 90% of the discordant pairs in both $M(I)$ and $M(I')$ had depth differences below 0.5 m, with ZoeDepth showing the highest proportion ($> 96\%$). Among the violation cases, more than 70% of ZoeDepth's discordant pairs differed by only 0.1 m, again the highest proportion among the evaluated models.

These results suggested that ZoeDepth's high ordinal inconsistency is mainly associated with pixel pairs having small depth differences, which may be less critical in many applications. This behavior may be related to its classification-regression architecture [24], which constrains depth estimates and reduces the sensitivity of individual depth values, but does not ensure the preservation of pairwise ordering. Although the current metric *Ord* does not exclude nearly equal pairs, it captures a different aspect of prediction consistency from *Diff*; both are informative for robustness evaluation.

The models showed different relative robustness characteristics across perturbations. For instance, UniDepthV2 generally showed lower MR violations under the *snow* perturbation but higher violations under the *bright* perturbation compared to ZoeDepth. ZoeDepth achieved perfect prediction consistency under horizontally flipped inputs, resulting in a 0% MR-violation rate. This behavior aligns with ZoeDepth's inference pipeline, which applies horizontal flip augmentation by default. Consistent with existing studies [6], [9], MDE models showed different robustness characteristics resulting from variations in architecture, training data, and optimization strategies.

Figs. 5 and 6 illustrate the MR-violation percentages measured by *Diff* and *Ord* in increasing perturbation severities. A strong positive correlation existed between the severity and MR-violation percentage, indicating progressively unstable predictions as input interventions intensify. The perturbation severity therefore serves as a parameter to adjust the difficulty of the robustness testing. Safety-critical applications with stringent reliability requirements can adopt higher perturbation severities to emulate challenging operational conditions, while less critical applications may rely on lower severities.

Under all perturbation severities, Depth Pro produced the highest MR-violation percentages under *Diff*, indicating the highest pixelwise sensitivity to input perturbations. ZoeDepth generally produced the lowest MR-violation percentages, indicating strong pixelwise consistency over a broad severity range. However, under noise corruptions (*Gaussian*, *impulse*, *shot*, and *ISO*), ZoeDepth's MR-violation percentage under *Diff* increased significantly beyond severity level 3, exceeding that of UniDepthV2 at level 4 and Depth Anything 3 at level 5. This suggested strong pixelwise consistency under low-noise

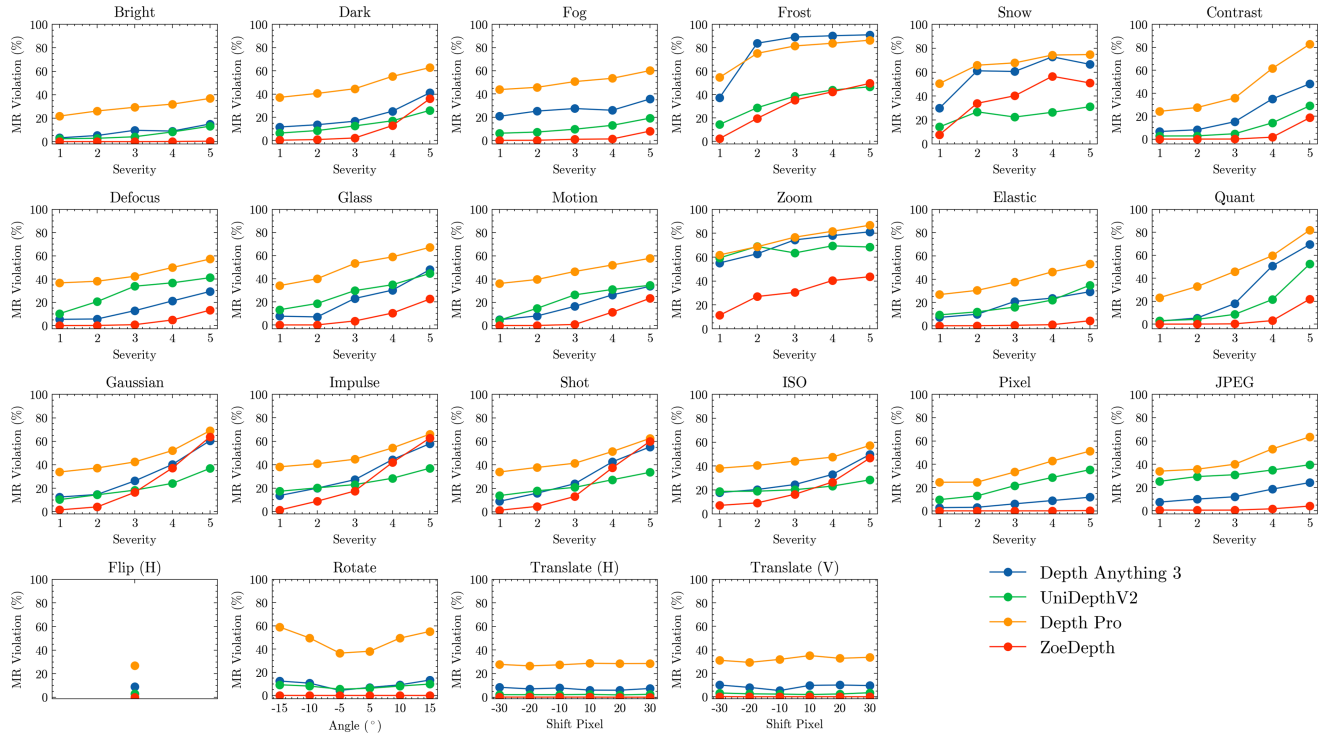


Fig. 5. MR violation (%) measured by Diff for four MDE models in four datasets with varying perturbation levels. Eighteen corruptions were evaluated at five severity levels, and four Euclidean transformations, except *horizontal flip*, were evaluated at six parameter settings.

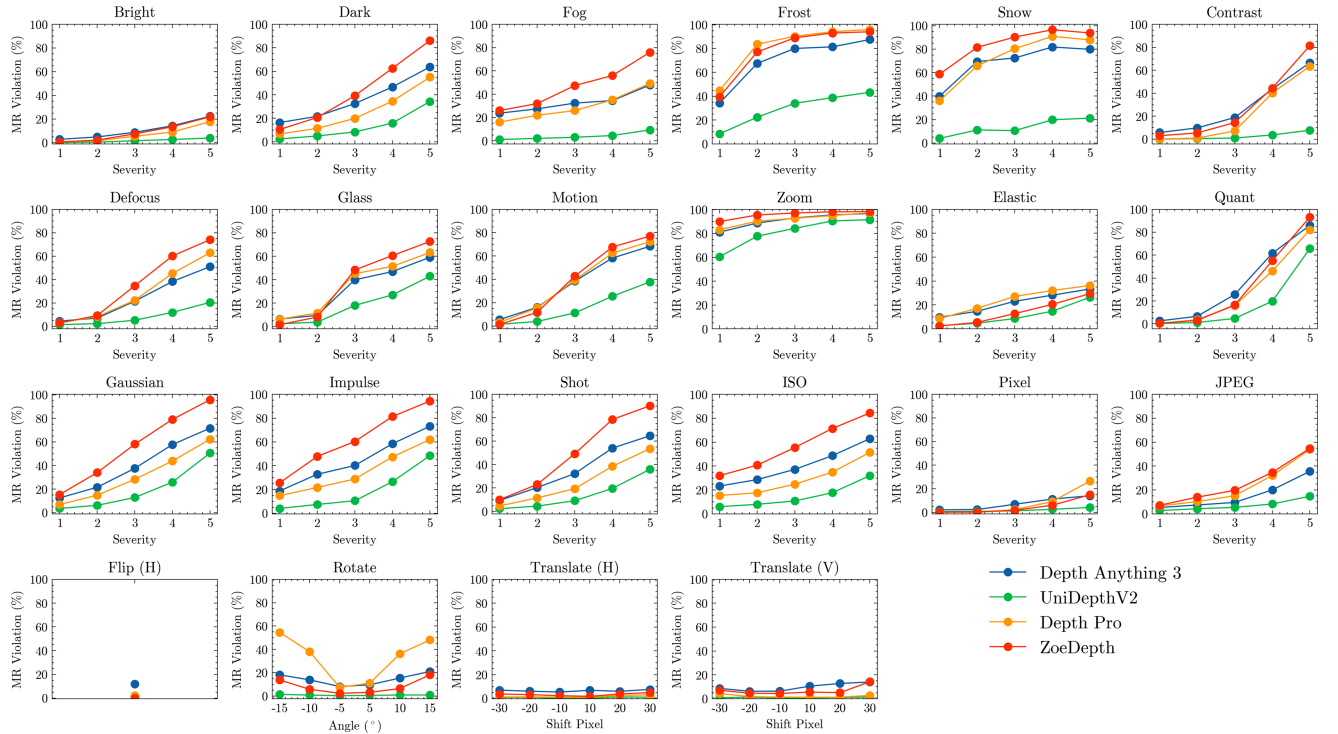


Fig. 6. MR violation (%) measured by Ord for four MDE models in four datasets with varying perturbation levels. Eighteen corruptions were evaluated at five severity levels, and four Euclidean transformations, except *horizontal flip*, were evaluated at six parameter settings.

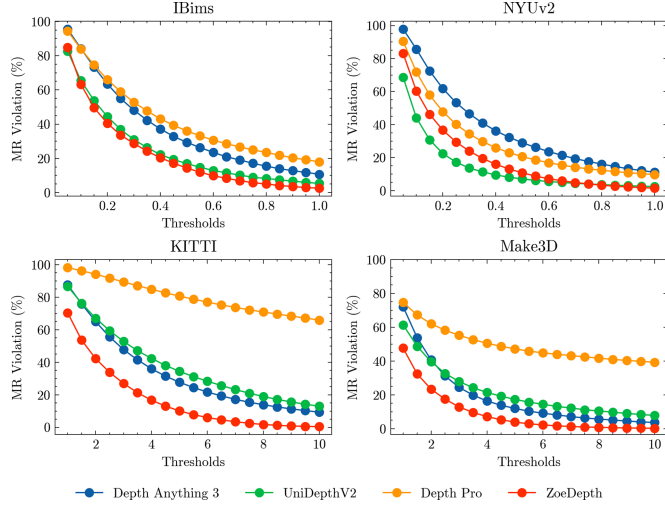


Fig. 7. MR violation (%) measured by Diff in four datasets with varying tolerance thresholds τ .

conditions but reduced stability under severe noise. By contrast, UniDepthV2 maintained relatively stable MR-violation percentages below Diff over severity levels, indicating greater robustness to severe noise. Despite its strong pixelwise consistency, ZoeDepth produced the highest MR-violation percentages under Ord, indicating the weakest ordinal consistency. UniDepthV2 consistently produced the lowest MR-violation percentages under Ord, indicating the strongest ordinal consistency over perturbation severities.

For Euclidean transformations, MR-violation percentages remained largely invariant across perturbation degrees. The selected parameter ranges for rotations and translations likely produce only minor visual differences between degrees, limiting their impact on prediction consistency. For example, horizontal translations of 10 and 20 pixels produce visually similar inputs. A minor exception was observed for Depth Pro under *vertical translation*, where the MR-violation percentages under Diff were slightly higher for downward shifts (shift pixel > 0) than for upward shifts (shift pixel < 0). Under *rotation*, clockwise and anticlockwise rotations yielded nearly identical violation percentages.

B. Tolerance Thresholds Selection

A sensitivity analysis was conducted to examine the effect of tolerance values on MR-violation percentages. Figs. 7 and 8 show the results for thresholds τ and θ . For Diff, τ was varied from 0.05 to 1.0 in increments of 0.05 for indoor datasets, and from 1.0 to 10.0 in increments of 0.5 for outdoor datasets to account for larger depth ranges. As expected, increasing τ consistently reduced MR-violation percentages under Diff: small τ values produced very high violation rates, while large τ values produced very low rates. For Ord, θ ranged from 0.99 to 0.90 in decreases of 0.005. Decreasing θ consistently reduced MR-violation percentages under Ord.

Over the tested τ range, Depth Pro produced the highest MR-violation percentages under Diff, except NYUv2, which

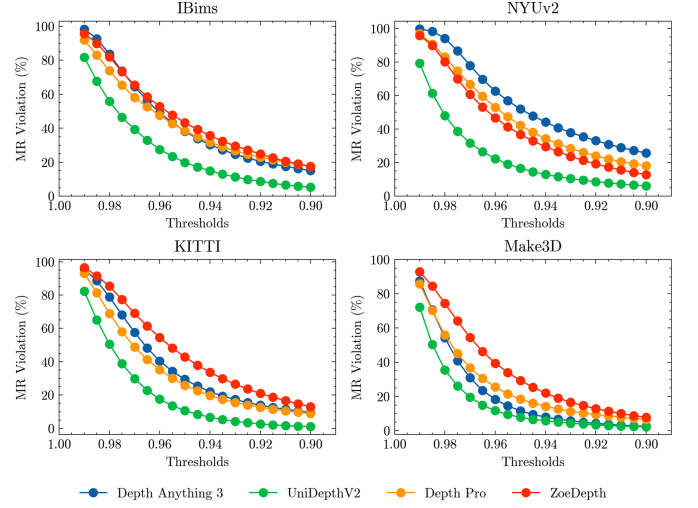


Fig. 8. MR violation (%) measured by Ord in four datasets with varying tolerance thresholds θ .

indicated the highest pixelwise sensitivity to perturbations. ZoeDepth produced the lowest MR-violation percentages under Diff, indicating the strongest pixelwise consistency. In contrast, ZoeDepth generally showed the highest MR-violation percentages under Ord over the tested θ range, whereas UniDepthV2 consistently showed the lowest. These model rankings remained stable over the tested tolerance ranges, indicating that the main conclusions are not sensitive to the choice of tolerance.

Although absolute tolerances were used in this study, relative tolerances based on predicted depth provide a viable alternative, since depth estimation errors (DEEs) often increase with distance. Such tolerances permit larger prediction differences for distant objects than for nearby objects and may be preferable in applications that place greater importance on nearby objects.

C. Comparisons of Robustness-Evaluation Metrics

Three commonly used robustness metrics that rely on ground-truth depth maps were calculated for comparison with MR-violation percentages. The DEE [9] integrates the absolute relative error (AbsRel) and threshold accuracy (δ_1), defined as

$$\text{DEE} = \frac{\text{AbsRel} - \delta_1 + 1}{2}. \quad (18)$$

The corruption error (CE) [9], [29] normalizes performance degradation under perturbations using ZoeDepth as the baseline, defined as

$$\text{CE} = \frac{\sum_{l=1}^L (\text{DEE}_l)}{\sum_{l=1}^L (\text{DEE}_l^{\text{baseline}})} \quad (19)$$

where L denotes the perturbation degree level. The resilience rate (RR) [9] measures the accuracy retained relative to clean inputs, defined as

$$\text{RR} = \frac{\sum_{l=1}^L (1 - \text{DEE}_l)}{L \times (1 - \text{DEE}_{\text{clean}})} \quad (20)$$

TABLE V
EVALUATION RESULTS IN THE IBIMS DATASET. FOUR MDE MODELS WERE EVALUATED: ZOEDDEPTH (ZOED), DEPTH PRO (DPRO), UNIDEPHTV2 (UNI2), AND DEPTH ANYTHING 3 (DA3)

Perturbation	DEE ↓				CE (%) ↓				RR (%) ↑				MR Viol. - Diff (%) ↓				MR Viol. - Ord (%) ↓			
	ZoeD	DPro	Uni2	DA3	ZoeD	DPro	Uni2	DA3	ZoeD	DPro	Uni2	DA3	ZoeD	DPro	Uni2	DA3	ZoeD	DPro	Uni2	DA3
Bright	0.244	0.174	0.083	0.138	100.00	71.23	34.12	56.60	97.83	99.62	99.25	99.40	0.00	4.80	1.60	8.40	12.40	6.80	4.40	14.00
Dark	0.307	0.228	0.107	0.218	100.00	74.40	34.82	70.97	89.67	93.03	96.69	90.20	15.60	26.80	14.60	26.40	45.60	28.60	21.60	39.60
Fog	0.276	0.299	0.103	0.150	100.00	108.40	37.24	54.28	93.73	84.54	97.15	98.07	4.40	32.20	4.20	18.80	50.00	38.00	6.20	23.60
Frost	0.387	0.648	0.223	0.607	100.00	167.31	57.52	156.89	79.32	42.50	84.16	45.30	37.60	82.60	30.60	80.80	83.00	88.00	40.40	74.00
Snow	0.404	0.490	0.169	0.370	100.00	121.37	41.83	91.48	77.13	61.46	89.97	72.71	45.40	75.00	24.20	52.00	89.00	76.60	19.40	70.00
Contrast	0.281	0.367	0.086	0.168	100.00	130.68	30.66	59.89	93.08	76.35	98.95	95.94	6.00	38.20	4.80	16.40	28.00	33.80	3.80	23.80
Defocus	0.278	0.242	0.119	0.158	100.00	86.90	42.73	56.93	93.42	91.44	95.40	97.07	7.80	32.60	18.40	18.00	43.60	35.20	16.60	29.00
Glass	0.288	0.352	0.154	0.247	100.00	122.51	53.57	85.82	92.18	78.08	91.58	86.86	9.60	48.60	17.80	32.20	48.80	57.00	30.60	40.00
Motion	0.286	0.240	0.090	0.164	100.00	83.87	31.48	57.44	92.42	91.67	98.53	96.39	10.00	34.60	7.20	16.40	46.00	41.20	22.00	42.00
Zoom	0.400	0.546	0.357	0.379	100.00	136.40	89.16	94.63	77.59	54.73	69.62	71.63	45.20	77.00	52.60	56.80	92.80	89.60	80.20	90.60
Elastic	0.268	0.212	0.212	0.246	100.00	79.06	79.36	91.87	94.78	95.07	85.28	86.98	2.60	27.00	33.00	33.60	28.60	26.60	16.40	28.00
Quant	0.277	0.299	0.136	0.272	100.00	108.11	49.04	98.15	93.57	84.48	93.56	83.98	9.00	37.80	15.60	35.80	40.60	38.60	22.80	47.20
Gaussian	0.356	0.263	0.121	0.295	100.00	73.98	34.03	82.91	83.34	88.82	95.15	81.29	29.00	31.80	17.00	43.40	64.20	42.80	32.00	57.00
Impulse	0.354	0.259	0.125	0.298	100.00	73.27	35.24	84.24	83.65	89.34	94.78	80.98	30.40	35.80	18.20	44.00	65.00	47.00	29.80	58.20
Shot	0.349	0.232	0.118	0.281	100.00	66.54	33.68	80.52	84.25	92.58	95.54	82.92	27.40	31.80	15.40	37.00	57.00	35.20	25.60	53.20
ISO	0.352	0.237	0.117	0.281	100.00	67.38	33.38	79.88	83.88	92.00	95.55	82.92	29.60	32.60	17.40	40.60	64.80	40.80	27.80	55.40
Pixel	0.232	0.285	0.198	0.135	100.00	123.03	85.37	58.14	99.43	86.22	86.86	99.80	0.20	29.60	38.00	8.20	7.80	11.20	5.60	10.40
JPEG	0.252	0.239	0.180	0.215	100.00	94.64	71.44	85.10	96.74	91.77	88.74	90.56	2.80	36.60	37.20	31.20	32.80	37.40	15.40	23.40
Flip (H)	0.227	0.173	0.082	0.166	100.00	76.09	35.96	73.16	100.00	99.73	99.42	96.16	0.00	5.00	0.00	14.00	0.00	5.00	3.00	16.00
Rotate	0.235	0.403	0.097	0.175	100.00	171.73	41.24	74.58	99.01	71.93	97.78	95.12	0.00	43.00	0.33	12.00	12.50	45.00	2.83	17.17
Translate (H)	0.229	0.203	0.077	0.133	100.00	88.49	33.59	58.24	99.76	96.14	99.93	99.94	0.00	6.33	0.00	7.83	5.33	3.50	2.83	10.50
Translate (V)	0.227	0.199	0.084	0.154	100.00	87.61	36.93	68.04	100.05	96.61	99.19	97.53	0.00	8.33	0.67	11.33	18.67	6.33	3.50	14.67
Mean	0.296	0.300	0.138	0.239	100.00	100.59	46.47	78.17	91.13	84.46	93.32	87.81	14.21	35.37	16.76	29.33	42.57	37.92	19.67	38.08
Source	0.227	0.171	0.076	0.133	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

The "mean" row reports the average over the 22 perturbations, and the "source" row reports DEE on the original images. For each metric, the best result for each perturbation is shown in bold.

TABLE VI
EVALUATION RESULTS IN THE NYUV2 DATASET

Perturbation	DEE ↓				CE (%) ↓				RR (%) ↑				MR Viol. - Diff (%) ↓				MR Viol. - Ord (%) ↓			
	ZoeD	DPro	Uni2	DA3	ZoeD	DPro	Uni2	DA3	ZoeD	DPro	Uni2	DA3	ZoeD	DPro	Uni2	DA3	ZoeD	DPro	Uni2	DA3
Bright	0.116	0.105	0.063	0.123	100.00	90.31	53.82	106.03	100.55	97.17	99.17	96.56	0.28	2.60	0.52	5.10	9.34	14.00	1.60	22.44
Dark	0.125	0.153	0.069	0.185	100.00	122.50	54.91	148.25	99.56	91.94	98.52	89.72	10.40	17.26	1.48	19.74	39.36	40.60	13.68	50.50
Fog	0.132	0.167	0.069	0.315	100.00	126.51	52.68	238.83	98.78	90.46	98.44	75.45	3.34	15.34	0.76	52.40	40.58	40.32	5.06	74.20
Frost	0.201	0.620	0.248	0.538	100.00	307.89	122.98	266.94	90.87	41.25	79.59	50.93	30.90	73.16	30.18	73.50	76.04	91.32	52.40	90.80
Snow	0.201	0.371	0.104	0.482	100.00	184.25	51.45	239.50	90.89	68.32	94.84	57.06	37.70	52.00	3.74	76.32	83.66	87.88	20.14	94.08
Contrast	0.108	0.229	0.054	0.231	100.00	212.55	49.95	214.95	101.55	83.75	100.11	84.68	3.64	21.32	0.34	31.46	18.98	28.64	1.04	49.94
Defocus	0.128	0.150	0.073	0.131	100.00	117.89	57.39	102.94	99.27	92.23	98.04	95.67	5.70	14.80	1.80	8.90	44.54	43.16	8.22	49.98
Glass	0.133	0.187	0.093	0.199	100.00	141.44	70.04	150.10	98.71	88.22	95.97	88.23	8.68	26.14	7.26	25.08	50.34	57.98	25.16	64.46
Motion	0.137	0.171	0.068	0.154	100.00	124.43	49.41	112.70	98.19	90.06	98.63	93.13	7.66	13.44	2.66	16.56	54.38	53.68	16.88	63.70
Zoom	0.208	0.507	0.377	0.430	100.00	243.45	181.01	206.55	90.10	53.54	65.92	62.78	43.00	65.62	62.66	68.14	97.64	99.10	85.06	99.90
Elastic	0.125	0.215	0.138	0.231	100.00	172.52	110.91	185.51	99.61	85.23	91.17	84.69	1.74	22.70	17.88	28.56	27.44	69.72	28.86	57.56
Quant	0.143	0.240	0.099	0.456	100.00	167.57	68.77	318.07	97.48	82.49	95.36	59.91	7.42	25.70	7.84	43.00	34.54	37.70	23.72	52.60
Gaussian	0.137	0.175	0.076	0.317	100.00	127.21	55.06	230.75	98.18	89.62	97.80	75.27	20.62	18.30	3.22	39.96	50.00	45.64	21.24	66.26
Impulse	0.136	0.170	0.075	0.326	100.00	124.78	54.93	239.22	98.28	90.10	97.87	74.22	20.08	17.78	3.14	43.64	52.68	53.18	21.34	75.56
Shot	0.123	0.150	0.066	0.317	100.00	122.20	53.70	257.52	99.80	92.26	98.81	75.28	16.62	13.66	2.30	42.36	42.62	42.32	15.46	65.28
ISO	0.121	0.148	0.067	0.295	100.00	121.94	55.07	243.77	100.01	92.54	98.74	77.63	16.82	12.50	1.80	39.16	48.14	47.64	15.82	74.04
Pixel	0.141	0.086	0.099	0.110	100.00	61.02	70.54	77.85	97.77	99.24	95.28	98.07	0.00	1.16	2.86	7.04	3.52	9.04	0.96	17.96
JPEG	0.134	0.165	0.091	0.140	100.00	122.52	68.02	104.10	98.51	90.70	96.12	94.74	0.16	21.66	2.90	4.70	15.88	28.80	5.28	22.08
Flip (H)	0.121	0.080	0.057	0.095	100.00	65.78	46.84	78.02	100.00	99.91	99.78	99.73	0.00	0.20	0.20	1.50	0.00	3.80	0.90	17.40
Rotate	0.248	0.261	0.177	0.211	100.00	105.12	71.24	85.31	85.58	80.28	87.11	86.85	0.00	10.97	0.22	2.97	4.02	21.90	0.35	14.83
Translate (H)	0.121	0.077	0.057	0.091	100.00	63.73	47.26	74.89	100.00	100.18	99.73	100.14	0.00	0.08	0.13	1.32	0.95	1.58	0.28	7.77
Translate (V)	0.115	0.078	0.057	0.087	100.00	68.06	50.05	76.01	100.73	100.09	99.71	100.53	0.03	0.07	0.03	0.98	5.67	1.28	0.33	9.00
Mean	0.143	0.205	0.103	0.248	100.00	136.08	68.00	170.81	97.47	86.34	94.85	82.78	10.67	20.29	7.00	28.74	36.38	41.79	16.54	51.83
Source	0.121	0.079	0.055	0.092	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Four MDE models were evaluated: ZoeDepth (ZoeD), Depth Pro (DPro), UniDepthV2 (Uni2), and Depth Anything 3 (DA3). The "mean" row reports the average over the 22 perturbations, and the "source" row reports DEE on the original images. For each metric, the best result for each perturbation is shown in bold.

where DEE_{clean} denotes the DEE measured in the source images. The calculation of RR assumes that DEE and DEE_{clean} are in the range [0, 1], which is not true for some perturbations and models in the KITTI and Make3D datasets. Therefore, for these two datasets, the DEE values were normalized to [0, 1] before computing RR by dividing each value by the maximum DEE (DEE_{max}) observed in all the models and perturbations for that dataset.

TABLE VII
EVALUATION RESULTS IN THE KITTI DATASET

Perturbation	DEE ↓				CE (%) ↓				RR (%) ↑				MR Viol. - Diff (%) ↓				MR Viol. - Ord (%) ↓			
	ZoeD	DPro	Uni2	DA3	ZoeD	DPro	Uni2	DA3	ZoeD	DPro	Uni2	DA3	ZoeD	DPro	Uni2	DA3	ZoeD	DPro	Uni2	DA3
Bright	0.302	0.210	0.108	0.309	100.00	69.51	35.89	102.61	100.51	99.16	99.22	100.83	0.24	68.78	10.74	14.32	5.80	2.44	0.48	3.06
Dark	0.262	0.357	0.100	0.430	100.00	136.49	38.04	164.14	101.21	96.63	99.37	98.70	12.14	87.12	27.62	33.84	56.92	22.52	12.54	45.06
Fog	0.313	0.273	0.138	0.383	100.00	86.99	44.07	122.35	100.30	98.09	98.72	99.53	0.32	84.52	19.98	23.84	45.10	7.06	1.44	22.26
Frost	0.303	0.463	0.143	0.781	100.00	152.68	47.23	257.39	100.47	94.82	98.63	92.48	34.96	91.52	49.04	81.86	83.38	74.58	14.46	78.56
Snow	0.326	0.455	0.142	0.504	100.00	139.85	43.70	154.79	100.09	94.96	98.65	97.39	55.06	91.00	39.22	70.84	97.28	72.18	5.06	85.14
Contrast	0.306	0.298	0.168	0.426	100.00	97.49	54.91	139.38	100.43	97.65	98.21	98.76	3.52	71.14	20.84	31.78	33.52	2.48	0.38	26.42
Defocus	0.396	0.264	0.609	0.391	100.00	66.63	153.74	98.63	98.84	98.23	90.78	99.39	1.68	89.64	75.74	26.18	43.80	28.34	5.56	16.06
Glass	0.358	0.279	0.470	0.417	100.00	77.90	131.20	116.64	99.51	97.98	93.13	98.92	10.88	87.84	74.82	31.30	48.62	24.40	16.22	23.02
Motion	0.326	0.345	0.398	0.363	100.00	105.76	122.09	111.41	100.08	96.85	94.34	99.88	11.10	91.50	63.92	33.48	48.48	44.76	19.70	35.08
Zoom	0.343	0.437	0.291	0.710	100.00	127.29	84.97	207.08	99.78	95.28	96.14	93.73	20.80	96.70	85.08	80.22	96.52	94.08	76.82	90.02
Elastic	0.318	0.190	0.075	0.401	100.00	59.76	23.63	126.19	100.22	99.50	99.78	99.21	0.16	66.78	8.56	8.30	0.14	0.72	0.38	0.80
Quant	0.347	0.236	0.110	0.481	100.00	68.03	31.84	138.64	99.71	98.71	99.18	97.80	3.46	79.36	28.70	27.28	34.76	25.66	17.92	32.24
Gaussian	0.260	0.331	0.087	0.392	100.00	127.36	33.42	150.86	101.24	97.08	99.58	99.36	32.82	85.20	44.68	32.10	68.14	27.72	17.62	34.72
Impulse	0.269	0.324	0.088	0.401	100.00	120.06	32.75	148.81	101.07	97.21	99.56	99.21	38.86	86.62	55.74	36.32	78.62	29.50	17.68	40.96
Shot	0.250	0.327	0.092	0.387	100.00	130.75	36.58	154.45	101.41	97.15	99.50	99.46	30.50	85.78	51.52	29.42	54.92	16.06	9.02	23.16
ISO	0.241	0.340	0.084	0.387	100.00	141.24	35.07	160.63	101.58	96.93	99.62	99.47	24.22	85.62	47.32	30.80	69.54	20.16	10.22	27.42
Pixel	0.372	0.318	0.323	0.335	100.00	85.55	86.95	90.03	99.27	97.31	95.60	100.39	0.26	76.40	40.14	6.74	7.34	10.36	0.88	0.64
JPEG	0.369	0.330	0.478	0.378	100.00	89.53	129.55	102.37	99.32	97.10	93.00	99.63	2.98	80.56	71.68	18.24	43.56	20.70	4.86	14.00
Flip (H)	0.330	0.152	0.058	0.373	100.00	46.15	17.47	112.83	100.00	100.14	100.07	99.71	0.00	66.10	2.50	18.60	0.00	0.20	0.00	8.40
Rotate	5.042	4.075	4.176	3.327	100.00	80.81	82.81	65.98	16.89	32.97	30.72	47.36	0.22	91.08	10.32	18.52	10.98	32.68	0.13	17.22
Translate (H)	0.342	0.184	0.056	0.301	100.00	53.65	16.24	88.01	99.79	99.61	100.11	100.98	0.15	66.10	0.07	15.95	0.82	0.28	0.03	3.97
Translate (V)	0.302	0.232	0.058	0.274	100.00	76.85	19.21	90.71	100.50	98.78	100.07	101.46	0.23	78.60	1.28	20.07	2.30	0.43	0.03	11.92
Mean	0.531	0.474	0.375	0.552	100.00	97.29	59.15	132.00	96.47	94.64	94.73	96.53	12.93	82.18	37.70	31.36	42.30	25.33	10.52	29.10
Source	0.330	0.161	0.062	0.357	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Four MDE models were evaluated: ZoeDepth (ZoeD), Depth Pro (DPro), UniDepthV2 (Uni2), and Depth Anything 3 (DA3). The “Mean” row reports the average over the 22 perturbations, and the “Source” row reports DEE on the original images. For each metric, the best result for each perturbation is shown in bold. To compute RR when DEE exceeds 1, the DEE values were normalized by the maximum DEE ($DEE_{max} = 6.0$) observed on KITTI over all models and perturbations.

TABLE VIII
EVALUATION RESULTS IN THE MAKE3D DATASET

Perturbation	DEE ↓				CE (%) ↓				RR (%) ↑				MR Viol. - Diff (%) ↓				MR Viol. - Ord (%) ↓			
	ZoeD	DPro	Uni2	DA3	ZoeD	DPro	Uni2	DA3	ZoeD	DPro	Uni2	DA3	ZoeD	DPro	Uni2	DA3	ZoeD	DPro	Uni2	DA3
Bright	0.732	1.887	0.323	0.408	100.00	257.67	44.16	55.69	99.98	123.47	100.16	99.47	0.00	40.74	12.02	6.02	8.88	4.22	0.60	2.86
Dark	0.714	5.270	0.333	0.435	100.00	738.22	46.62	60.98	100.27	41.78	100.02	99.06	3.62	61.06	12.78	6.64	33.40	10.38	4.98	9.62
Fog	0.769	1.528	0.347	0.443	100.00	198.65	45.08	57.63	99.39	132.13	99.81	98.94	0.46	71.12	19.38	13.24	54.28	33.38	3.48	12.78
Frost	0.707	0.832	0.322	0.741	100.00	117.73	45.63	104.79	100.39	148.95	100.17	94.45	15.80	58.04	27.97	77.14	71.86	73.08	10.22	37.30
Snow	0.663	1.337	0.323	0.448	100.00	201.75	48.73	67.57	101.08	136.74	100.16	98.87	13.24	48.44	29.48	33.54	66.78	52.20	8.86	25.56
Contrast	0.742	2.694	0.376	0.423	100.00	363.23	50.67	57.05	99.83	103.99	99.37	99.24	3.32	54.88	16.82	10.96	38.66	24.68	5.26	15.64
Defocus	0.779	3.377	0.357	0.352	100.00	433.78	45.87	45.20	99.24	87.48	99.65	100.32	0.00	42.56	18.34	6.46	12.76	6.62	2.56	2.72
Glass	0.761	3.177	0.366	0.356	100.00	417.28	48.12	46.80	99.51	92.32	99.51	100.25	0.00	40.14	12.94	3.92	5.28	2.72	3.48	2.12
Motion	0.764	3.543	0.350	0.369	100.00	463.66	45.73	48.23	99.47	83.47	99.77	100.06	0.00	46.34	15.48	5.58	11.56	14.58	5.28	8.44
Zoom	0.731	3.513	0.530	0.655	100.00	480.48	72.44	89.63	100.00	84.21	97.07	95.74	13.38	61.20	63.46	76.10	96.68	84.36	81.80	83.90
Elastic	0.729	2.158	0.364	0.379	100.00	296.02	49.98	52.01	100.03	116.92	99.54	99.90	0.00	38.78	15.94	2.26	0.16	0.00	0.00	0.78
Quant	0.733	2.802	0.369	0.436	100.00	382.42	50.39	59.54	99.97	101.37	99.47	99.04	0.46	51.88	19.70	10.84	24.98	16.70	9.04	14.00
Gaussian	0.662	3.496	0.334	0.431	100.00	528.19	50.49	65.15	101.10	84.62	100.00	99.12	16.10	52.46	18.50	7.98	43.44	8.74	8.42	2.86
Impulse	0.662	3.531	0.323	0.422	100.00	533.51	48.76	63.79	101.10	83.76	100.17	99.25	16.56	55.04	23.76	6.62	51.28	10.22	8.88	4.06
Shot	0.657	3.493	0.324	0.435	100.00	531.31	49.25	66.23	101.17	84.70	100.15	99.06	18.78	50.38	21.64	8.58	46.32	8.58	7.06	2.88
ISO	0.657	3.755	0.323	0.428	100.00	571.45	49.12	65.19	101.18	78.36	100.17	99.16	14.12	51.44	21.18	5.70	44.84	6.76	5.74	3.78
Pixel	0.746	2.444	0.338	0.377	100.00	327.56	45.24	50.50	99.76	110.03	99.95	99.94	0.00	34.28	5.72	4.20	0.00	0.00	0.16	0.46
JPEG	0.729	2.803	0.343	0.377	100.00	384.75	47.10	51.81	100.04	101.35	99.86	99.93	0.00	42.26	16.38	3.76	12.02	7.52	2.86	2.88
Flip (H)	0.732	2.340	0.335	0.370	100.00	319.72	45.83	50.52	99.99	112.54	99.98	100.05	0.00	36.10	9.00	2.30	0.00	0.00	0.00	6.00
Rotate	0.749	0.853	0.321	0.415	100.00	113.84	42.86	55.48	99.71	148.45	100.20	99.36	0.00	46.75	20.57	4.53	5.67	30.70	0.00	8.38
Translate (H)	0.736	2.127	0.337	0.396	100.00	289.07	45.84	53.77	99.92	117.67	99.95	99.65	0.00	39.32	8.53	3.27	6.17	1.53	0.00	3.80
Translate (V)	0.730	1.459	0.337	0.387	100.00	199.89	46.21	53.04	100.01	133.80	99.95	99.78	0.00	42.25	8.65	2.78	0.65	0.40	0.00	3.40
Mean	0.722	2.655	0.349	0.431	100.00	370.46	48.37	60.03	100.14	104.91	99.78	99.12	5.27	48.43	19.01	13.75	28.89	18.06	7.67	11.56
Source	0.731	2.859	0.334	0.373	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Four MDE models were evaluated: ZoeDepth (ZoeD), Depth Pro (DPro), UniDepthV2 (Uni2), and Depth Anything 3 (DA3). The “Mean” row reports the average over the 22 perturbations, and the “Source” row reports DEE on the original images. For each metric, the best result for each perturbation is shown in bold. To compute RR when DEE exceeds 1, the DEE values were normalized by the maximum DEE ($DEE_{max} = 7.0$) observed on Make3D over all models and perturbations.

under Diff than ZoeDepth. For perturbations, such as *zoom blur*

TABLE IX
p-VALUES FROM TWO-SIDED WILCOXON SIGNED-RANK TESTS FOR EACH
EVALUATION METRIC

Dataset	Metric	ZoeD versus DPro	ZoeD versus Uni2	ZoeD versus DA3	Uni2 versus DPro	Uni2 versus DA3	DPro versus DA3
IBims	DEE	<u>0.2705</u>	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
	CE	<u>0.1523</u>	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
	RR	<0.0001	0.0025	0.0003	<0.0001	<0.0001	<u>0.0107</u>
	Diff	<0.0001	<u>0.0902</u>	<0.0001	<0.0001	<0.0001	0.0004
	Ord	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<u>0.2894</u>
NYUv2	DEE	0.0003	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
	CE	0.0004	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
	RR	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<u>0.0167</u>
	Diff	<0.0001	<u>0.1423</u>	<0.0001	<0.0001	<0.0001	<0.0001
	Ord	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
KITTI	DEE	<u>0.0442</u>	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
	CE	<u>0.5128</u>	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
	RR	<0.0001	<0.0001	0.0003	<u>0.0742</u>	0.0003	<0.0001
	Diff	<0.0001	<0.0001	<0.0001	<0.0001	<u>0.0382</u>	<0.0001
	Ord	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<u>0.0024</u>
Make3D	DEE	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
	CE	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
	RR	<u>0.0342</u>	0.0011	<0.0001	<u>0.0236</u>	<0.0001	<u>0.0198</u>
	Diff	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
	Ord	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001

Six pairwise comparisons are performed among four models, ZoeDepth (ZoeD), Depth Pro (DPro), UniDepthV2 (Uni2), and Depth Anything 3 (DA3), using DEE, CE, RR, and MR-violation percentage (measured by Diff and Ord). The results with $p > \alpha = 0.0083$ are underlined, indicating that there were no statistically significant differences between the corresponding pair of models.

match the image dimensions, fine-grained depth details were smoothed. Consequently, small perturbation-induced changes in predicted depth did not substantially affect the measured error relative to the ground truth, resulting in similarly high RR values across models.

The MR-violation percentages indicated that ZoeDepth produced the strongest pixelwise prediction consistency but the weakest pairwise ordinal depth consistency on all datasets except NYUv2, with generally statistically significant differences compared to the other three models (typically $p < 0.0001$). This tradeoff is likely due to frequent order reversals in pixel pairs with similar depth values, all of which are counted as discordant by Ord. The discrepancy between the results of the RR and the MR-violation under Diff in most of the datasets further indicates that the stability of the accuracy does not necessarily imply prediction consistency, consistent with [15].

Input perturbations generally reduced the prediction accuracy, as indicated by increases in DEE for perturbed images relative to source images. However, cases with $RR > 100$ indicate that certain perturbations led to improved accuracy. For example, on the Make3D dataset, ZoeDepth exhibited a substantial decrease in DEE under noise corruption. The corresponding MR-violation percentages under Diff remained relatively low (14%–18%), whereas those under Ord were higher (43%–52%) than for most other models. This indicated stable pixelwise predictions but weaker pairwise ordinal consistency, despite improved accuracy. In the same dataset, Depth Pro showed significant DEE reductions under *frost* and *rotation*, but also high MR-violation percentages (58% under Diff and 73% under Ord for *frost*; 47% under Diff and 31% under Ord for *rotation*), indicating inconsistent predictions despite improved accuracy. On KITTI, *rotation* produced $DEE > 1$ for all models, likely due to large errors in distant regions, such as the sky. Nevertheless, all

TABLE X
COMPARISON OF ROBUSTNESS-EVALUATION METRICS

Metric	Requires GT	Relation to Accuracy	Uses Source Prediction As Reference
DEE	✓	Direct	×
CE	✓	Indirect	×
RR	✓	Indirect	✓
MR Violation	×	None	✓

models except Depth Pro showed relatively low MR-violation percentages (below 20%) under this perturbation, indicating consistent predictions for the rotation angles tested (1° – 3°). In general, these results demonstrate that the MR-violation percentage provides insights that complement conventional robustness-evaluation metrics.

Table X compares the key characteristics of the commonly used robustness-evaluation metrics with the MR-violation percentage. Unlike conventional metrics, the evaluation of the MR-violation percentage does not require comparisons with ground-truth depth data, thereby fully decoupling robustness evaluation from accuracy assessment. In contrast, DEE directly measures prediction accuracy by quantifying the error of the estimated depths under perturbations. CE measures the relative degradation of the accuracy caused by perturbations, whereas RR measures the stability of the accuracy. Therefore, both CE and RR assess robustness indirectly through changes in prediction accuracy under input perturbations.

Compared with existing robustness-evaluation metrics, the primary advantage of MRT4Depth lies in its ability to evaluate MDE models without requiring ground-truth depth data. This capability enables robustness assessment across a broader range of datasets and proves particularly beneficial in scenarios where acquiring dense and reliable depth annotations incurs high cost.

MRT4Depth directly quantifies prediction consistency under perturbations and provides insights that complement the accuracy-based evaluations. However, high prediction consistency does not necessarily imply high accuracy. Thus, relying solely on MRT4Depth may lead to biased conclusions. For instance, ZoeDepth exhibited perfect prediction consistency under horizontal flip perturbations but produced higher DEE values than UniDepthV2, indicating lower accuracy for that perturbation. When ground-truth depth data are available, MRT4Depth serves as a valuable complement by revealing stability characteristics that conventional metrics do not capture. In the absence of ground truth, a high MR-violation percentage indicates inconsistent predictions and suggests a lack of robustness, whereas a low MR-violation percentage indicates relatively consistent predictions without guaranteeing high accuracy.

D. Comparison of Robustness Testing Approaches

Table XI summarizes the main characteristics of MRT4Depth, representative MDE robustness benchmarks, and adversarial robustness testing methods. Unlike benchmarks, such as RoboDepth [9] and PDE [15], MRT4Depth does not require ground-truth depth and evaluates prediction consistency rather than accuracy or accuracy stability. These benchmark-based

TABLE XI
COMPARISON OF ROBUSTNESS TESTING APPROACHES

Approach	Requires GT	Tested Perturbations	Evaluation Criterion
RoboDepth [9]	✓	Image corruption	Accuracy, accuracy stability
PDE [15]	✓	3D perturbation	Accuracy, accuracy stability, prediction consistency
SAAM [43], 3D ² Fool [44]	×	Adversarial patch insertion	Vulnerability
MRT4Depth (Ours)	×	Image corruption & Euclidean transformation	Prediction consistency

methods become difficult to apply when ground truth is unavailable. Adversarial testing methods, such as SAAM [43] and 3D²Fool [44], also operate without ground truth, but focus on model's vulnerability to intentionally designed perturbations. In contrast, MRT4Depth evaluates consistency under image perturbations that may occur naturally.

E. Limitations

Since MRT4Depth evaluates prediction consistency rather than accuracy, its robustness assessment is not comprehensive. Prediction consistency does not imply prediction correctness. For example, a trivial model that always outputs a constant depth map would achieve perfect consistency while remaining entirely inaccurate. MRT4Depth should therefore be interpreted as a measure of model stability that complements accuracy-based metrics. A high MR-violation percentage provides strong evidence of inconsistent predictions and limited robustness under nonadversarial perturbations, even when accuracy is high. In contrast, a low MR-violation percentage indicates relatively consistent predictions but does not guarantee correctness unless supported by accuracy-based evaluation.

In addition, Ord is computed using Kendall's Tau, which does not exclude pixel pairs with similar depth values. As a result, many such pairs are counted as discordant, reducing the ordinal consistency score.

The Euclidean-transformation design constrains follow-up images to contain only a small fraction of invalid pixels. Although this fraction is below 10% of the image, invalid pixels can still affect MDE predictions. Furthermore, combinations of multiple perturbations that better reflect real operating conditions are not considered.

Moreover, this study is limited to model-level evaluation rather than system-level evaluation. Assessment of downstream depth-based applications, such as monocular SLAM, 3-D reconstruction, and visual odometry, could provide further insight into the relationship between MR-violation percentage and practical system reliability. However, such analysis would require integration of MDE models into complete pipelines and control of additional factors, including camera intrinsics and pose estimation, which is beyond the scope of this article.

Finally, this study considers only metric depth estimation models and excludes affine-invariant models, which require alignment with ground-truth depth to determine scale and shift. This setting is outside the scope of this work.

V. CONCLUSION

This study introduces MRT4Depth, a ground-truth-free robustness-evaluation method for MDE models. The method addresses the oracle problem in robustness assessment when depth data are unavailable and enables evaluation on unlabeled datasets. Instead of measuring prediction accuracy, MRT4Depth identifies MR violations by directly quantifying prediction consistency under input perturbations using pixelwise depth difference and pairwise ordinal consistency, thus providing insights that complement existing MDE robustness-evaluation metrics. Experiments conducted on four MDE datasets showed that the four evaluated state-of-the-art MDE models exhibited different strengths and weaknesses in maintaining consistent depth predictions across 22 input perturbations. The results indicated that Depth Pro produced the weakest pixelwise prediction consistency, whereas ZoeDepth produced the strongest pixelwise consistency, but the weakest pairwise ordinal consistency. UniDepthV2 produced the strongest pairwise ordinal consistency and the highest prediction accuracy. The results also showed that prediction accuracy and accuracy stability do not necessarily imply prediction consistency. These findings support a deeper understanding of the reliability of MDE model predictions under diverse real-world deployment conditions.

Future work includes formulating additional MRs to assess robustness under broader conditions, including changes in scene geometry or camera viewpoint. The effects of combined perturbations also merit exploration. Another possible direction is to replace the absolute tolerance thresholds with relative ones. Further study will also consider real-world video data, where source and follow-up images are drawn from consecutive frames rather than simulated transformations. In addition, examining the relationship between robustness measured by MRT4Depth and downstream task performance under nonadversarial perturbations remains an important direction. Beyond evaluation, integrating MRT into training pipelines may support the development of robustness-aware MDE models.

REFERENCES

- [1] Z. Huang et al., "A systematic review of monocular depth estimation for autonomous driving: Methods and dataset benchmarking," *Results Eng.*, vol. 26, 2025, Art. no. 105359.
- [2] X. Dong, M. A. Garratt, S. G. Anavatti, and H. A. Abbass, "Towards real-time monocular depth estimation for robotics: A survey," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 10, pp. 16940–16961, Oct. 2022.
- [3] H. Zheng, S. Rajadnya, and A. Zakhori, "Monocular depth estimation for drone obstacle avoidance in indoor environments," in *Proc. 2024 IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2024, pp. 10027–10034.
- [4] X. Wang, B. Yang, M. Wei, L. Liu, J. Zhang, and Y. Nie, "Deep learning for endoscopic depth estimation: A review," *Displays*, vol. 90, 2025, Art. no. 103086.
- [5] N. Drenkow, N. Sani, I. Shpitser, and M. Unberath, "A systematic review of robustness in deep learning for computer vision: Mind the gap?," 2022, *arXiv:2112.00639*.

- [6] K. Xian, Z. Cao, C. Shen, and L. Guosheng, "Towards robust monocular depth estimation: A new baseline and benchmark," *Int. J. Comput. Vis.*, vol. 132, no. 7, pp. 2401–2419, 2024.
- [7] B. Li et al., "Trustworthy AI: From principles to practices," *ACM Comput. Surv.*, vol. 55, no. 9, pp. 1–46, 2023.
- [8] S. Gasperini, N. Morbitzer, H. Jung, N. Navab, and F. Tombari, "Robust monocular depth estimation under challenging conditions," in *Proc. 2023 IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 8143–8152.
- [9] L. Kong, S. Xie, H. Hu, L. X. Ng, B. R. Cottreau, and W. T. Ooi, "RoboDepth: Robust out-of-distribution depth estimation under corruptions," *Adv. Neural Inf. Process. Syst.*, vol. 36, pp. 21298–21342, 2023.
- [10] R. Sharifian, N. Rabbani, and A. Bartoli, "The RoDEM benchmark: Evaluating the robustness of monocular single-shot depth estimation methods in minimally-invasive surgery," *Int. J. Comput. Assist. Radiol. Surg.*, vol. 20, no. 6, pp. 1215–1229, 2025.
- [11] Z. Cheng, C. Han, J. Liang, Q. Wang, X. Zhang, and D. Liu, "Self-supervised adversarial training of monocular depth estimation against physical-world attacks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 9084–9101, Dec. 2024.
- [12] T. Freiesleben and T. Grote, "Beyond generalization: A theory of robustness in machine learning," *Synthese*, vol. 202, no. 4, pp. 1–28, 2023.
- [13] U. Rajapaksha, F. Sohel, H. Laga, D. Diepeveen, and M. Bennamoun, "Deep learning-based depth estimation methods from monocular image and videos: A comprehensive survey," *ACM Comput. Surv.*, vol. 56, no. 12, pp. 1–51, 2024.
- [14] E. T. Barr, M. Harman, P. McMinn, M. Shahbaz, and S. Yoo, "The oracle problem in software testing: A survey," *IEEE Trans. Softw. Eng.*, vol. 41, no. 5, pp. 507–525, May 2015.
- [15] J. Nugent et al., "Evaluating robustness of monocular depth estimation with procedural scene perturbations," 2025, *arXiv:2507.00981*.
- [16] Z. Song et al., "ER-Depth: Enhancing the robustness of self-supervised monocular depth estimation in challenging scenes," *ACM Trans. Multimedia Comput., Commun. Appl.*, vol. 21, no. 12, pp. 1–23, 2025.
- [17] S. Segura, D. Towey, Z. Q. Zhou, and T. Y. Chen, "Metamorphic testing: Testing the untestable," *IEEE Softw.*, vol. 37, no. 3, pp. 46–53, May/Jun. 2020.
- [18] T. Y. Chen et al., "Metamorphic Testing: A review of challenges and opportunities," *ACM Comput. Surv.*, vol. 51, no. 1, pp. 1–27, 2018.
- [19] Z. Q. Zhou, T. H. Tse, and M. Witheridge, "Metamorphic robustness testing: Exposing hidden defects in citation statistics and journal impact factors," *IEEE Trans. Softw. Eng.*, vol. 47, no. 6, pp. 1164–1183, Jun. 2021.
- [20] J. Wang, Y. Ishimoto, M. Kondo, Y. Kamei, and N. Ubayashi, "Uncertainty-aware metamorphic testing for robust object detection models," in *Proc. 2023 IEEE 23rd Int. Conf. Softw. Qual., Rel., Secur. Companion*, 2023, pp. 9–17.
- [21] Z. Hou, X. Wang, S. Zhang, and Z. Chen, "SegTest: Metamorphic testing of image segmentation via guided instance-level test data augmentation," *Softw. Testing, Verification Rel.*, vol. 35, no. 1, 2025, Art. no. e1910.
- [22] D. Eigen, C. Puhrsch, and R. Fergus, "Depth map prediction from a single image using a multi-scale deep network," in *Proc. 28th Int. Conf. Neural Inf. Process. Syst.*, 2014, pp. 2366–2374.
- [23] R. Ranftl, A. Bochkovskiy, and V. Koltun, "Vision transformers for dense prediction," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 12179–12188.
- [24] S. F. Bhat, R. Birkel, D. Wofk, P. Wonka, and M. Müller, "ZoeDepth: Zero-shot transfer by combining relative and metric depth," 2023, *arXiv:2302.12288*.
- [25] A. Bochkovskii et al., "Depth Pro: Sharp monocular metric depth in less than a second," in *Proc. Int. Conf. Learn. Representations*, 2025, pp. 1–36.
- [26] L. Piccinelli et al., "UniDepthV2: Universal monocular metric depth estimation made simpler," 2025, *arXiv:2502.20110*.
- [27] H. Lin et al., "Depth Anything 3: Recovering the visual space from any views," 2025, *arXiv:2511.10647*.
- [28] Z. Xu et al., "Towards depth foundation model: Recent trends in vision-based depth estimation," in *Proc. Comput. Vis. Media*, vol. 12, no. 2, Apr. 2025, pp. 243–271.
- [29] D. Hendrycks and T. G. Dietterich, "Benchmarking neural network robustness to common corruptions and perturbations," in *Proc. Int. Conf. Learn. Representations*, 2019, pp. 1–16.
- [30] A. Saxena, M. Sun, and A. Y. Ng, "Make3D: Learning 3D scene structure from a single still image," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 31, no. 5, pp. 824–840, May 2009.
- [31] T. Y. Chen, S. C. Cheung, and S. M. Yiu, "Metamorphic Testing: A new approach for generating next test cases," Dept. Comput. Sci., Hong Kong Univ. Sci. Technol., Hong Kong, Tech. Rep. HKUST-CS98-01, 1998.
- [32] F. T. Chan, T. Y. Chen, S. C. Cheung, M. F. Lau, and S. M. Yiu, "Application of metamorphic testing in numerical analysis," in *Proc. IASTED Int. Conf. Softw. Eng.*, 1998, pp. 191–197.
- [33] R. Li et al., "Metamorphic relation generation: State of the art and research directions," *ACM Trans. Softw. Eng. Methodol.*, vol. 34, no. 5, pp. 1–25, 2025.
- [34] S. Wang and Z. Su, "Metamorphic object insertion for testing object detection systems," in *Proc. 35th IEEE/ACM Int. Conf. Automated Softw. Eng.*, 2021, pp. 1053–1065.
- [35] M. Duran, T. Laurent, E. Rushe, and A. Ventresque, "Metamorphic testing for pose estimation systems," in *Proc. 2025 IEEE Conf. Softw. Testing, Verification Validation*, 2025, pp. 47–57.
- [36] Y. Tian, K. Pei, S. Jana, and B. Ray, "DeepTest: Automated testing of deep-neural-network-driven autonomous cars," in *Proc. IEEE/ACM 40th Int. Conf. Softw. Eng.*, 2018, pp. 303–314.
- [37] Y. Zhang et al., "Metamorphic testing harness for the Baidu Apollo perception-camera module," in *Proc. IEEE/ACM 8th Int. Workshop Metamorphic Testing*, 2023, pp. 9–16.
- [38] T. Van Dijk and G. De Croon, "How do neural networks see depth in single images?," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 2183–2191.
- [39] T. Koch, L. Liebel, F. Fraundorfer, and M. Körner, "Evaluation of CNN-based single-image depth estimation methods," in *Proc. Eur. Conf. Comput. Vis. Workshops*, 2019, pp. 331–348.
- [40] Y. Hu et al., "Out-of-distribution monocular depth estimation with local invariant regression," *Knowl.-Based Syst.*, vol. 319, 2025, Art. no. 113518.
- [41] P. K. N. Silberman, D. Hoiem, and R. Fergus, "Indoor segmentation and support inference from RGBD images," in *Proc. Eur. Conf. Comput. Vis.*, 2012, pp. 746–760.
- [42] J. Uhrig, N. Schneider, L. Schneider, U. Franke, T. Brox, and A. Geiger, "Sparsity invariant CNNs," in *Proc. Int. Conf. 3D Vis.*, 2017, pp. 11–20.
- [43] A. Guesmi, M. A. Hanif, B. Ouni, and M. Shafique, "SAAM: Stealthy adversarial attack on monocular depth estimation," *IEEE Access*, vol. 12, pp. 13571–13585, 2024.
- [44] J. Zheng, C. Lin, J. Sun, Z. Zhao, Q. Li, and C. Shen, "Physical 3D adversarial attacks against monocular depth estimation in autonomous driving," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 24452–24461.